

Calculus through Economic Applications

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Outline

- ① Single-Variable Calculus through Firms' Profit Maximization Problems
- ② Taylor Approximation through Risk Premium
- ③ Multi-Variable Calculus through Unconstrained Optimization
- ④ Integration through Normal Distributions
- ⑤ Leibniz Formula through Continuation Values (Optional)

Firm's Decision Making

- We review single-variable calculus to solve the following problems.

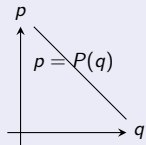
Profit Maximization Problems

- 1 Consider the profit maximization problem of a **price-taking** firm:

$$\max_{q \in [0, \infty)} \underbrace{p \cdot q}_{\text{revenue}} - \underbrace{c(q)}_{\text{cost}}.$$

- 2 Consider the profit maximization problem of a **price-setting** firm:

$$\max_{q \in [0, \infty)} P(q) \cdot q - c(q),$$



where **P is the inverse demand function.**

The cost function c is differentiable, strictly increasing, strictly convex, and satisfies $c(0) = c'(0) = 0$ and $\lim_{q \rightarrow \infty} c(q) = \lim_{q \rightarrow \infty} c'(q) = \infty$.

Also, assume that $P(q) \cdot q$ is concave (e.g., $P(q) = a - bq$).

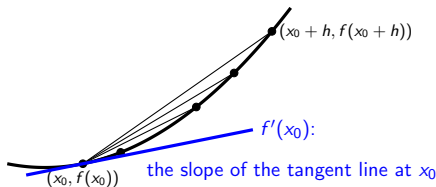
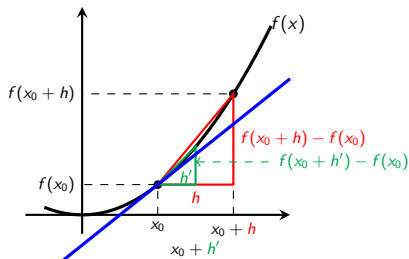
Derivatives

Definition 3.1

- ① Let $f : (a, b) \rightarrow \mathbb{R}$ be a function, where $-\infty \leq a < b \leq \infty$.
The function f is *differentiable* at a point $x_0 \in (a, b)$ if

$$f'(x_0) := \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \text{ exists and is finite.}$$

- ② We call the limit $f'(x_0)$ (also denoted $Df(x_0)$, $\frac{df}{dx}(x_0)$) the *derivative of f at x_0* (if it exists and is finite).



First-Order Condition

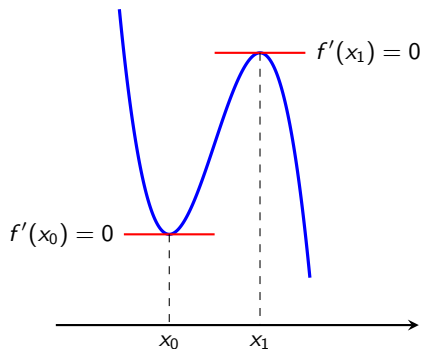
First-Order Condition

- Let $f : A \subseteq \mathbb{R} \rightarrow \mathbb{R}$.
- Suppose that x_0 is an **interior** point of A and that f is **differentiable** at x_0 .
- **If** x_0 is a local **maximizer** or **minimizer** of f (i.e., x_0 is a local extremal point of f) on A , **then**

$$f'(x_0) = 0.$$

- The above condition is referred to as the **first-order (necessary) condition (FOC, FONC)**.
- Such x_0 (x_0 with $f'(x_0) = 0$) is called a stationary point/a critical point.

First-Order Condition: Intuition



- Suppose that f is differentiable at an interior point x .
- If $f'(x) > 0$, then
a small increase/decrease in x increases/decreases the value of f .
- If $f'(x) < 0$, then
a small decrease/increase in x increases/decreases the value of f .

First-Order Condition for Concave/Convex Functions

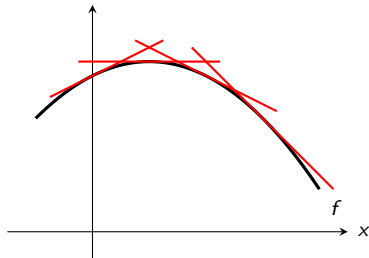
Concave Maximization or Convex Minimization

Let $f : (a, b) \rightarrow \mathbb{R}$ be a $\begin{cases} \text{concave} \\ \text{convex} \end{cases}$ and differentiable function.

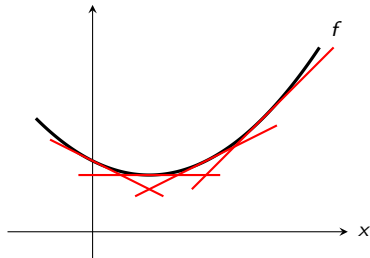
A point $x_0 \in (a, b)$ is a global $\begin{cases} \text{maximizer} \\ \text{minimizer} \end{cases}$ of f on (a, b)

if and only if $f'(x_0) = 0$.

(Strictly) Concave



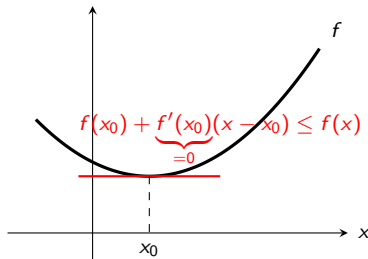
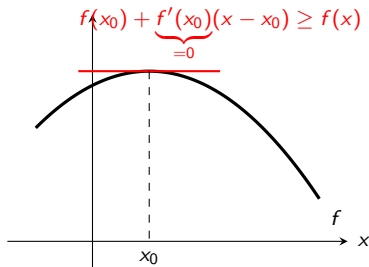
(Strictly) Convex



First-Order Condition for Concave/Convex Functions

Another Intuition

The tangent lines always lie above/below its graph:



- So, we want to take the derivative of $p \cdot q - c(q)$ and $P(q) \cdot q - c(q)$.

Properties of Derivatives

Proposition 3.3

Let $f, g : (a, b) \rightarrow \mathbb{R}$ be two differentiable functions at $x \in (a, b)$.

For any linear combination $\alpha f + \beta g : (a, b) \rightarrow \mathbb{R}$ with $\alpha, \beta \in \mathbb{R}$, we have

$$(\alpha f + \beta g)'(x) = \alpha \cdot f'(x) + \beta \cdot g'(x).$$

In particular,

$$\begin{cases} (f + g)'(x) = f'(x) + g'(x) \\ (\alpha f)' = \alpha \cdot f'(x) \end{cases}.$$

- ① We can exchange the order of **linear combinations** and **taking derivatives**.

$$\begin{aligned} \blacktriangleright f, g &\longrightarrow \alpha f + \beta g \longrightarrow (\alpha f + \beta g)' \\ \blacktriangleright f, g &\longrightarrow f', g' \longrightarrow \alpha f' + \beta g'. \end{aligned}$$

② $(p \cdot q - c(q))' = p \cdot \underbrace{q'}_{=1} - c'(q).$

Properties of Derivatives

Proof Sketch (Optional)

- Always try to recall $\left\{ \begin{array}{l} \text{what you want/have to show} \\ \text{(key) definitions/results} \end{array} \right.$

$$(\alpha f + \beta g)'(x)$$

|| def. of derivatives

$$\lim_{h \rightarrow 0} \frac{(\alpha f + \beta g)(x+h) - (\alpha f + \beta g)(x)}{h}$$

|| def. of the sum of functions

$$\lim_{h \rightarrow 0} \frac{((\alpha f)(x+h) + (\beta g)(x+h)) - ((\alpha f)(x) + (\beta g)(x))}{h}$$

|| def. of the scalar product of a function

$$\lim_{h \rightarrow 0} \frac{(\alpha \cdot f(x+h) + \beta \cdot g(x+h)) - (\alpha \cdot f(x) + \beta \cdot g(x))}{h}$$

WTS
(What To Show)

$$\alpha f'(x) + \beta g'(x)$$

|| def. of derivatives

$$\alpha \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} + \beta \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h}$$

|| "algebra of limits"

$$\lim_{h \rightarrow 0} \left(\alpha \frac{f(x+h) - f(x)}{h} + \beta \frac{g(x+h) - g(x)}{h} \right)$$

|| "algebra"

$$\lim_{h \rightarrow 0} \frac{\alpha(f(x+h) - f(x)) + \beta(g(x+h) - g(x))}{h}$$

"algebra"

Properties of Derivatives

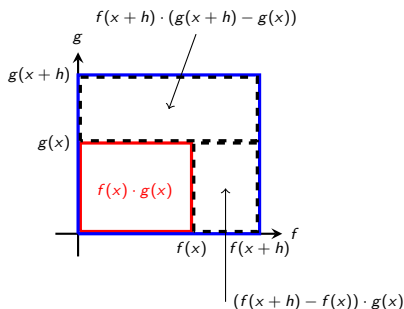
Proposition 3.4

Let $f, g : (a, b) \rightarrow \mathbb{R}$ be differentiable at $x \in (a, b)$.

Then, $fg : (a, b) \rightarrow \mathbb{R}$ is differentiable at $x \in (a, b)$ and

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x).$$

Proof Sketch



$$\begin{aligned} & \frac{f(x+h)g(x+h) - f(x)g(x)}{h} \\ &= \frac{f(x+h) - f(x)}{h} g(x) + f(x+h) \frac{g(x+h) - g(x)}{h} \\ &\rightarrow f'(x)g(x) + f(x)g'(x) \text{ (as } h \rightarrow 0). \end{aligned}$$

Properties of Derivatives

Proposition 3.5

Let $f, g : (a, b) \rightarrow \mathbb{R}$ be differentiable at $x \in (a, b)$. Suppose further that $g(x) \neq 0$.

Then, $\frac{f}{g} : (a, b) \rightarrow \mathbb{R}$ is differentiable at $x \in (a, b)$ and

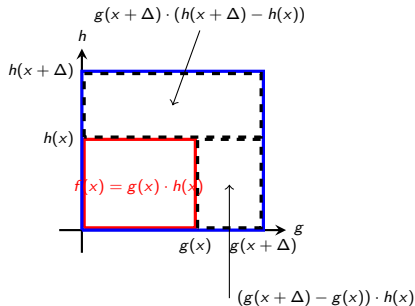
$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}.$$

Properties of Derivatives

- If you forget the quotient rule, remember how to derive it

Derivation #1

- 1) Let $h = \frac{f}{g}$. Then, $f = gh$.
- 2) By the product rule, $f' = (gh)' = g'h + gh'$.
- 3) So, $h' = \frac{f' - g'h}{g} = \frac{f' - g' \frac{f}{g}}{g} = \frac{f'g - fg'}{g^2}$.



$$\begin{aligned} f'(x) &\approx \frac{g(x + \Delta)h(x + \Delta) - g(x)h(x)}{\Delta} \\ &= \frac{g(x + \Delta) - g(x)}{\Delta} h(x) + g(x + \Delta) \frac{h(x + \Delta) - h(x)}{\Delta} \\ \rightarrow f'(x) &= g'(x)h(x) + g(x)h'(x). \end{aligned}$$

Properties of Derivatives

Derivation #2

$$\left(\frac{f}{g}\right)' = \left(f \cdot \frac{1}{g}\right)' = f' \frac{1}{g} + f \underbrace{\left(\frac{1}{g}\right)'}_{=-\frac{g'}{g^2}} = \frac{fg' - fg'}{g^2}.$$

$$\frac{\frac{1}{g(x+h)} - \frac{1}{g(x)}}{h} = \frac{\left(\frac{g(x) - g(x+h)}{g(x+h)g(x)}\right)}{h} = \frac{-\frac{g(x+h) - g(x)}{h}}{g(x+h)g(x)} \rightarrow -\frac{g'(x)}{(g(x))^2}.$$

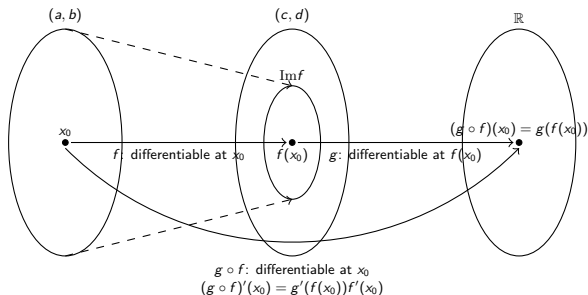
Properties of Derivatives

Proposition 3.6

Let $a, b, c, d \in \mathbb{R}$ be such that $-\infty \leq a < b \leq \infty$ and $-\infty \leq c < d \leq \infty$. Let $f : (a, b) \rightarrow \mathbb{R}$ and $g : (c, d) \rightarrow \mathbb{R}$ be functions such that $f((a, b)) \subseteq (c, d)$. Suppose that f is differentiable at $x_0 \in (a, b)$ and g is differentiable at $f(x_0) \in (c, d)$.

Then, $g \circ f$ is differentiable at $x_0 \in (a, b)$ and

$$(g \circ f)'(x_0) = g'(f(x_0)) \cdot f'(x_0).$$



Properties of Derivatives

Example 3.4

- Since $\left(\frac{1}{y}\right)' = -\frac{1}{y^2}$,

$$\left(\frac{1}{g(x)}\right)' = -\frac{1}{(g(x))^2}g'(x) = -\frac{g'(x)}{(g(x))^2}.$$

- Now,

$$\left(\frac{f}{g}\right)' = \left(f \cdot \frac{1}{g}\right)' = f' \frac{1}{g} - f \frac{g'}{g^2} = \frac{f'g - fg'}{g^2}.$$

Properties of Derivatives

Intuition

- $\underbrace{f(x) - f(x_0)}_{\text{change in } y} \approx f'(x_0) \underbrace{(x - x_0)}_{\text{change in } x}$ (if $x \approx x_0$).
- $\underbrace{g(y) - g(y_0)}_{\text{change in } z} \approx g'(y_0) \underbrace{(y - y_0)}_{\text{change in } y}$ (if $y \approx y_0$).
- Letting $y = f(x)$ and $y_0 = f(x_0)$,

$$\begin{aligned} g(f(x)) - g(f(x_0)) &\approx g'(f(x_0))(f(x) - f(x_0)) \\ &\approx \underbrace{g'(f(x_0))f'(x_0)}_{(g \circ f)'(x_0)} (x - x_0). \end{aligned}$$

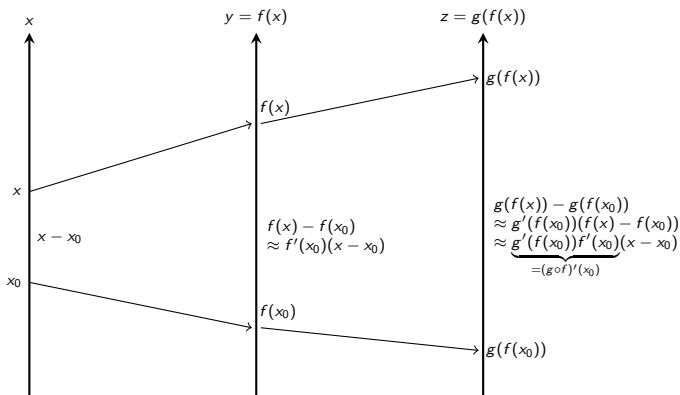
- In other words,

$$\begin{aligned} \frac{\underbrace{(g \circ f)(x) - (g \circ f)(x_0)}_{x - x_0}}{} &\approx g'(f(x_0))f'(x_0) \\ \implies (g \circ f)'(x_0) &= g'(f(x_0))f'(x_0). \end{aligned}$$

Properties of Derivatives

- This suggests:

$$\frac{dz}{dx} = \frac{dz}{dy} \frac{dy}{dx}.$$



Properties of Derivatives

- Demand function: $p \mapsto q = D(p)$
- Inverse demand function: $q \mapsto p = P(q)$
- If $y = f(x)$ and $g = f^{-1}$ then

$$\underbrace{(g \circ f)'(x)}_{=x'=1} = (f^{-1})'(f(x)) \cdot f'(x), \text{ i.e., } (f^{-1})'(f(x)) = \frac{1}{f'(x)}.$$

$$x \longmapsto f(x) \quad \longmapsto x = f^{-1}(f(x))$$

$$y = f(x) \quad \longmapsto f^{-1}(y) = x \quad \longmapsto y = f(x)$$

Properties of Derivatives

Derivative of an Inverse Function

$$(f^{-1})'(f(x)) = \frac{1}{f'(x)} \quad (f'(x) \neq 0)$$

$$\text{or } (f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))} \quad (f'(f^{-1}(y)) \neq 0).$$

$$f'(x) = \frac{dy}{dx}$$

$$(f^{-1})'(\underbrace{y}_{=f(x)}) = \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} = \frac{1}{f'(\underbrace{x}_{=f^{-1}(y)})}$$

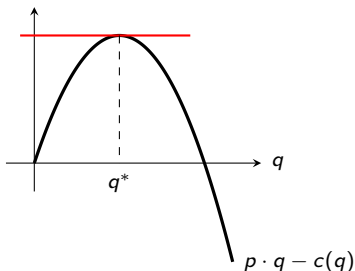
Price-Taking Firm

- Consider the profit maximization problem of a price-taking firm:

$$\max_{q \in [0, \infty)} p \cdot q - c(q).$$

- By taking the first-order condition with respect to q , the optimal quantity q^* is given by

$$\underbrace{c'(q^*)}_{\text{marginal cost}} = \underbrace{p}_{\text{price}} \quad \text{or} \quad q^* = (c')^{-1}(p).$$



Price-Setting Firm

- Consider the profit maximization problem of a price-setting firm:

$$\max_{q \in [0, \infty)} P(q) \cdot q - c(q).$$

- The first-order condition yields:

$$\underbrace{P'(q^*)q^*}_{\text{effect of } q \text{ on } P} + \underbrace{P(q^*)}_{\text{price}} = \underbrace{c'(q^*)}_{\text{marginal cost}},$$

that is,

$$\underbrace{P(q^*)}_{\text{price}} \left(1 + \frac{P'(q^*)q^*}{P(q^*)} \right) = \underbrace{c'(q^*)}_{\text{marginal cost}}.$$

Price-Setting Firm

- Now,

$$\frac{P'(q^*)q^*}{P(q^*)} = \frac{\frac{dP}{dq}q}{P} \bigg|_{q=q^*} = \frac{\frac{dP}{P}}{\frac{dq}{q}} \bigg|_{q=q^*}$$

is the ratio between the percentage change in price $\frac{dP}{P}$ and the percentage change in quantity $\frac{dq}{q}$.

- In terms of the demand function $q = D(P)$,

$$\frac{\frac{dP}{dq}q}{P} = \frac{1}{\frac{dq}{dP} \frac{P}{q}}.$$

Now, $\frac{dq}{dP} \frac{P}{q}$ is the ratio between the percentage change in quantity (demand) and that in price, that is, the price elasticity of demand.

Price-Setting Firm

- Denoting by $\eta := \frac{dq}{dP} \frac{P}{q} (< 0)$, the FOC is:

$$P(q^*) \left(1 + \frac{1}{\eta} \right) = c'(q^*).$$

- Thus, the price set by the (monopoly) firm satisfies:

$$P(q^*) = \underbrace{\left(\frac{1}{1 + \eta^{-1}} \right)}_{>1} c'(q^*) > c'(q^*).$$

- The term

$$\frac{1}{1 + \eta^{-1}} (> 1)$$

is referred to as the *markup* so that the *price* equals to the *markup* times the *marginal cost*.

Price-Setting Firm

Exercise 11.14 (Optional)

Consider the profit maximization problem of the price-setting firm. Suppose that the problem is given instead by:

$$\max_{p \in [0, \infty)} p \cdot D(p) - c(D(p)),$$

where D is the demand function and c is the cost function which satisfies the previously-mentioned properties. Assuming that the first-order condition characterizes the unique optimum, show that the optimal price p^* satisfies:

$$p^* = \left(\frac{1}{1 + \eta^{-1}} \right) c'(D(p^*)),$$

where η is the price elasticity of demand: $\eta = \frac{dD(p)}{dp} \frac{p}{D(p)} \Big|_{p=p^*}$.

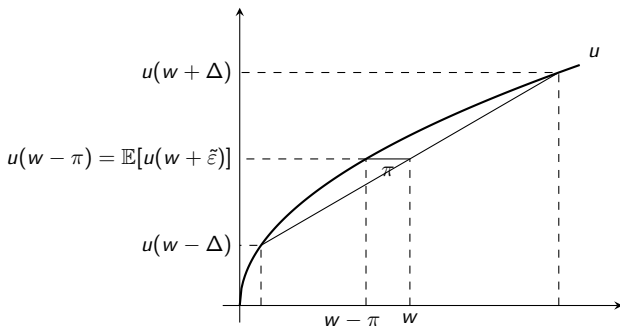
Risk Premium

- We review Taylor approximation through the following question.

Question

- Starting with wealth w , how much would a decision maker pay (π , risk premium) to avoid a gamble $\tilde{\varepsilon}$ with mean 0 and variance σ^2 ?

$$u(w - \pi) = \mathbb{E}[u(w + \tilde{\varepsilon})].$$



Risk Premium

Risk Premium

$$u(w - \pi) = \mathbb{E}[u(w + \tilde{\varepsilon})].$$

Overview

- The first-order approximation of the LHS is:

$$u(w - \pi) \approx u(w) - u'(w)\pi.$$

- The second-order approximation of the RHS is:

$$\mathbb{E}[u(w + \tilde{\varepsilon})] \approx u(w) + u'(w) \underbrace{\mathbb{E}[\tilde{\varepsilon}]}_{=0} + \frac{1}{2} u''(w) \underbrace{\mathbb{E}[\tilde{\varepsilon}^2]}_{=\sigma^2 \text{ given } \mathbb{E}[\tilde{\varepsilon}]=0}.$$

- Then, by algebra, the risk premium π is approximated by

$$\pi \approx \frac{1}{2} \underbrace{\left(-\frac{u''(w)}{u'(w)} \right)}_{\text{absolute risk aversion}} \sigma^2.$$

Taylor Expansion

We want to approximate a given function $\overset{u}{f}$ at a point $\overset{w}{x_0}$ by a polynomial.

$n = 1$: the linear approximation of f at x_0 by

$$\underbrace{f(x_0 + h)}_{\text{value of } f \text{ around } x_0} \approx \underbrace{\alpha_0 + \alpha_1 h}_{\text{"linear"}}.$$

$n = 2$: the quadratic approximation of f at x_0 by

$$f(x_0 + h) \approx \underbrace{\alpha_0 + \alpha_1 h + \alpha_2 h^2}_{\text{"quadratic"}}.$$

It turns out that:

$$f(x_0 + h) \approx f(x_0) + f'(x_0)h \text{ and}$$

$$f(x_0 + h) \approx f(x_0) + f'(x_0)h + \frac{f''(x_0)}{2}h^2.$$

($\Rightarrow$) How do we determine the coefficients $(\alpha_0, \alpha_1, \dots, \alpha_n)$?

Taylor Expansion

–1) We would like to approximate a function f at x_0 by a polynomial:

$$\underbrace{f(x_0 + h)}_{\substack{\text{value of } f \\ \text{around } x_0}} = \underbrace{\alpha_0 + \alpha_1 h + \alpha_2 h^2 + \cdots + \alpha_n h^n}_{\text{polynomial}} (+\text{something small}).$$

0) Letting $h = 0$ yields

$$\alpha_0 = f(x_0).$$

1) Differentiate both sides with respect to h to get

$$f'(x_0 + h) = \alpha_1 + 2\alpha_2 h + \cdots + n\alpha_n h^{n-1}.$$

Letting $h = 0$ yields

$$\alpha_1 = f'(x_0).$$

Taylor Expansion

2) Differentiate both sides with respect to h to get

$$f''(x_0 + h) = 2\alpha_2 + (3 \cdot 2)\alpha_3 h + \cdots + n(n-1)\alpha_n h^{n-2}.$$

Letting $h = 0$ yields

$$\alpha_2 = \frac{f''(x_0)}{2}.$$

3) Differentiate both sides with respect to h to get

$$f'''(x_0 + h) = (3 \cdot 2 \cdot 1)\alpha_3 + (4 \cdot 3 \cdot 2)\alpha_4 h + \cdots + n(n-1)(n-2)\alpha_n h^{n-3}.$$

Letting $h = 0$ yields

$$\alpha_3 = \frac{f^{(3)}(x_0)}{3!}.$$

$\vdots$

Taylor Expansion

$n - 1$) Differentiate both sides with respect to h to get

$$f^{(n-1)}(x_0 + h) = ((n-1)(n-2) \cdots 1)\alpha_{n-1} + (n(n-1)(n-2) \cdots 2)\alpha_n h.$$

Letting $h = 0$ yields

$$\alpha_{n-1} = \frac{f^{(n-1)}(x_0)}{(n-1)!}.$$

n) Differentiate both sides with respect to h to get

$$f^{(n)}(x_0 + h) = (n(n-1)(n-2) \cdots 1)\alpha_n, \text{ i.e., } \alpha_n = \frac{f^{(n)}(x_0)}{n!}.$$

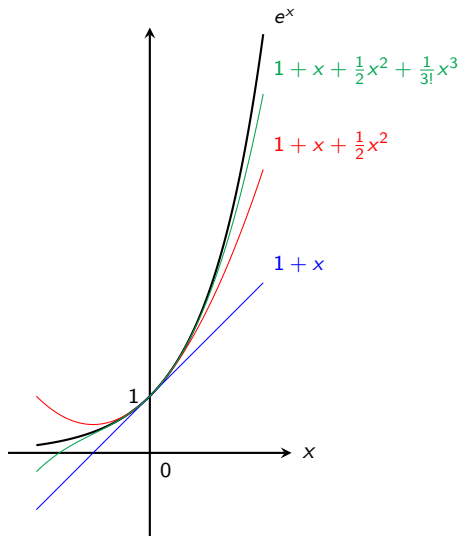
Taylor Expansion

Example: $f(x) = e^x$

- Consider $f(x) = e^x$. We have $f'(x) = e^x$.
 - ▶ $e \approx (1 + \frac{1}{n})^n$ (when n is “large”).
 - ▶ $e \approx (1 + x)^{\frac{1}{x}}$ (when $x \approx 0$).
 - ▶ $e^x \approx 1 + x$.
- Thus, $f'(x) = e^x, f''(x) = e^x, f^{(3)}(x) = e^x, \dots$
- Since $e^0 = 1$, the Taylor expansion of order $n \in \{1, 2, 3\}$ of f at $x_0 = 0$ is:

$$\begin{cases} e^x \approx 1 + x; \\ e^x \approx 1 + x + \frac{1}{2}x^2; \\ e^x \approx 1 + x + \frac{1}{2}x^2 + \frac{1}{3!}x^3. \end{cases}$$

Taylor Expansion

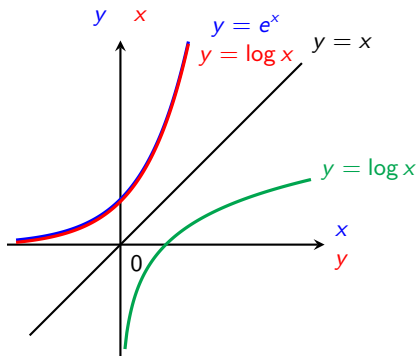


Taylor Expansion

Derivative of $y = \log x$: $(\log x)' = \frac{1}{x}$

- $y = e^x \Rightarrow \frac{dy}{dx} = y$

- $x = \log y \quad y = \log x$
 $\frac{dx}{dy} \stackrel{(?)}{=} \frac{1}{\frac{dy}{dx}} = \frac{1}{y} \implies \frac{dy}{dx} = \frac{1}{x}$



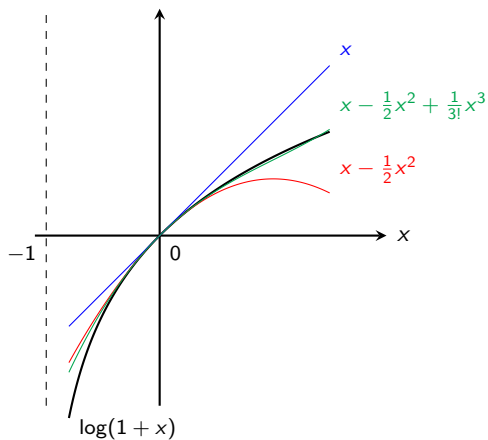
Taylor Expansion

Example: $f(x) = \log(1+x)$

- Consider $f(x) = \log(1+x)$.
- We have $(\log(1+x))' = \frac{1}{1+x}$
because $(\log x)' = \frac{1}{x}$.
- $f'(x) = \frac{1}{1+x}$, $f''(x) = -\frac{1}{(1+x)^2}$, $f^{(3)}(x) = 2\frac{1}{(1+x)^3}, \dots$
- Thus, $f'(0) = 1$, $f''(0) = -1$, $f^{(3)}(0) = 2, \dots$
- The Taylor expansion of order $n \in \{1, 2, 3\}$ of f at $x_0 = 0$ is

$$\begin{cases} \log(1+x) \approx x; \\ \log(1+x) \approx x - \frac{1}{2}x^2; \\ \log(1+x) \approx x - \frac{1}{2}x^2 + \frac{1}{3}x^3. \end{cases}$$

Taylor Expansion



Aside: An Application of $(\log x)' = \frac{1}{x}$

Example (Growth Accounting)

- $(\log f(x))' = \frac{f'(x)}{f(x)}.$
- Consider $Y(t) = A(t)K^\alpha(t)L^{1-\alpha}(t).$
- Then, $\log Y(t) = \log A(t) + \alpha \log K(t) + (1 - \alpha) \log L(t),$ and we have:

$$\frac{Y'(t)}{Y(t)} = \frac{A'(t)}{A(t)} + \alpha \frac{K'(t)}{K(t)} + (1 - \alpha) \frac{L'(t)}{L(t)}.$$

Aside: Applications of $\log(1 + x) = x$

Growth Rate

- Consider a sequence $(Y_t)_{t \in \mathbb{N}}$ of positive real numbers.
- At each (time) $t \geq 2$, denote its growth rate by $r_t = \frac{Y_t - Y_{t-1}}{Y_{t-1}}$.
- Then, the growth rate at time t can be approximated by

$$\Delta \log Y_t := \log Y_t - \log Y_{t-1}$$

because

$$\begin{aligned}\Delta \log Y_t &= \log Y_t - \log Y_{t-1} \\ &= \log \left(\frac{Y_t}{Y_{t-1}} \right) \\ &= \log \left(1 + \frac{Y_t - Y_{t-1}}{Y_{t-1}} \right) \\ &= \log(1 + r_t) \\ &\approx r_t.\end{aligned}$$

Aside: Applications of $\log(1 + x) = x$ (Optional)

Fisher Equation

- Denote by i is the nominal interest rate. Denote by π the inflation rate. Denote by r is the real interest rate. Then,

$$(1 + i) = (1 + \pi)(1 + r).$$

- Taking log,

$$\log(1 + i) = \log(1 + \pi) + \log(1 + r).$$

- Thus,

$$i \approx \log(1 + i) = \log(1 + \pi) + \log(1 + r) \approx \pi + r.$$

- Hence, as an approximation, we obtain Fisher equation:

$$i = \pi + r \text{ or } r = i - \pi.$$

Aside: Applications of $\log(1 + x) = x$

Log Linearization

- By the first-order approximation,

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0).$$

- Thus,

$$\begin{aligned} f(x) &\approx f(x_0) + f'(x_0) \underbrace{(x - x_0)}_{\text{absolute change}} \\ &= f(x_0) + f'(x_0)x_0 \cdot \underbrace{\frac{x - x_0}{x_0}}_{\text{percentage change}} \\ &\approx f(x_0) + f'(x_0)x_0 \cdot (\log x - \log x_0). \end{aligned}$$

Risk Premium

- $\tilde{\varepsilon}$ is a random variable with mean 0 and variance σ^2 .
- Letting π be the risk premium,

$$u(w - \pi) = \mathbb{E}[u(w + \tilde{\varepsilon})].$$

- On the one hand, for each realization ε , by the 2nd order Taylor approximation,

$$u(w + \varepsilon) \approx u(w) + u'(w)\varepsilon + \frac{1}{2}u''(w)\varepsilon^2.$$

- Taking the expectation, the RHS is approximated by

$$\mathbb{E}[u(w + \tilde{\varepsilon})] \approx u(w) + \underbrace{u'(w)\mathbb{E}[\tilde{\varepsilon}]}_{=0} + \frac{1}{2}u''(w) \underbrace{\mathbb{E}[\tilde{\varepsilon}^2]}_{=\sigma^2 \text{ given } \mathbb{E}[\tilde{\varepsilon}]=0}.$$

- On the other hand, the 1st order Taylor approximation of the LHS is:

$$u(w - \pi) \approx u(w) - u'(w)\pi.$$

- Then, the risk premium π is approximated by

$$\pi \approx \underbrace{\frac{1}{2} \left(-\frac{u''(w)}{u'(w)} \right)}_{\text{absolute risk aversion}} \sigma^2.$$

Risk Premium

- Remark on variance:

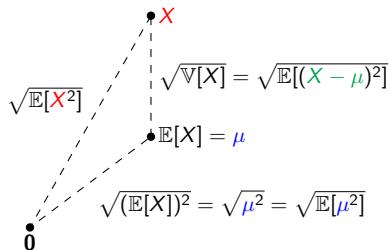
$$\underbrace{\mathbb{V}[\tilde{\varepsilon}]}_{=\sigma^2} = \mathbb{E}[\tilde{\varepsilon}^2] - \underbrace{(\mathbb{E}[\tilde{\varepsilon}])^2}_{=0}$$
$$\implies \sigma^2 = \mathbb{E}[\tilde{\varepsilon}^2]$$

Variance

$$\mathbb{V}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Proof:

$$\begin{aligned}\mathbb{V}[X] &= \mathbb{E}[(X - \mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2 - 2\mathbb{E}[X]X + (\mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2] - 2\mathbb{E}[X]\mathbb{E}[X] + (\mathbb{E}[X])^2 \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2.\end{aligned}$$



Unconstrained Optimization

- We quickly review multivariable calculus to solve unconstrained optimization problems.

Representative Firm's Problem (Optional Exercise)

Let $\Pi : \mathbb{R}_+^2 \rightarrow \mathbb{R}$ be defined by

$$\Pi(K, L) := AK^\alpha L^{1-\alpha} - (rK + wL) \text{ for each } (K, L) \in \mathbb{R}_+^2.$$

Consider:

$$\max_{K, L} \Pi(K, L).$$

Simple Linear Regression Problem

Given $(x_i, y_i)_{i=1}^n$, consider:

$$\min_{(\beta_1, \beta_2)} \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i)^2.$$

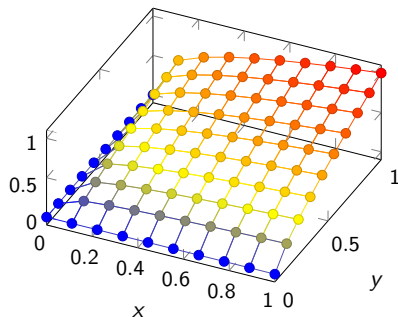
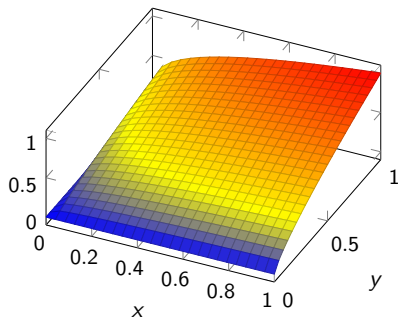
Partial Derivatives

The basic idea is

to analyze a function $f(x_1, x_2, \dots, x_n)$ variable-by-variable, coordinate-by-coordinate, one at a time for each variable.

Example

- $f(x, y) = x^{0.3}y^{0.7}$

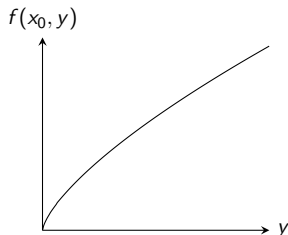


Partial Derivatives

- If we fix $x = x_0$, then

$$f(\underbrace{x = x_0}_{\text{fixed}}, y) = \underbrace{x_0^{0.3}}_{\text{constant}} y^{0.7}$$

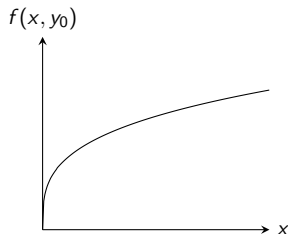
is a (power) function of y .



- If we fix $y = y_0$, then

$$f(x, \underbrace{y = y_0}_{\text{fixed}}) = \underbrace{y_0^{0.7}}_{\text{constant}} x^{0.3}$$

is a (power) function of x .



Partial Derivatives

- A partial derivative: a derivative with respect to **one** variable, while **keeping all the other variables constant**.
- The partial derivative of $f(x_1, x_2)$ with respect to x_1 :

$$\frac{\partial}{\partial x_1} f(x_1, x_2) := \lim_{h \rightarrow 0} \frac{f(x_1 + h, x_2) - f(x_1, x_2)}{h}.$$

- The partial derivative of $f(x_1, x_2)$ with respect to x_2 :

$$\frac{\partial}{\partial x_2} f(x_1, x_2) := \lim_{h \rightarrow 0} \frac{f(x_1, x_2 + h) - f(x_1, x_2)}{h}.$$

Partial Derivatives

Example

- $f(x, y) = x^\alpha y^{1-\alpha}$.
- The partial derivative of f with respect to x is

$$\frac{\partial}{\partial x} f(x, y) = (y^{1-\alpha})(\alpha x^{\alpha-1}) = \alpha x^{-(1-\alpha)} y^{1-\alpha}.$$

- The partial derivative of f with respect to y is

$$\frac{\partial}{\partial y} f(x, y) = (x^\alpha)((1-\alpha)y^{(1-\alpha)-1}) = (1-\alpha)x^\alpha y^{-\alpha}.$$

Partial Derivatives

Definition 3.5

- ① A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ admits *the partial derivatives at $\mathbf{x} = (x_1, \dots, x_n)$* if, for every $i = 1, \dots, n$, the limit
for each direction i

$$\frac{\partial f}{\partial x_i}(\mathbf{x}) := \lim_{h \rightarrow 0} \frac{f(x_1, \dots, x_{i-1}, \mathbf{x}_i + h, x_{i+1}, \dots, x_n) - f(x_1, \dots, x_{i-1}, \mathbf{x}_i, x_{i+1}, \dots, x_n)}{h}$$

exists and is finite.

- ② In this case, for each $i = 1, \dots, n$, we call it the *i -th partial derivative of f at $\mathbf{x}$* (the partial derivative of f with respect to x_i at $\mathbf{x}$) and denote it by

$$\frac{\partial f}{\partial x_i}(\mathbf{x}), \frac{\partial f(\mathbf{x})}{\partial x_i}, \frac{\partial}{\partial x_i} f(\mathbf{x}), f_{x_i}(\mathbf{x}), \text{ or } f'_{x_i}(\mathbf{x}).$$

- ③ Also, we call

$$\nabla f(\mathbf{x}) := \left(\frac{\partial f}{\partial x_1}(\mathbf{x}), \dots, \frac{\partial f}{\partial x_i}(\mathbf{x}), \dots, \frac{\partial f}{\partial x_n}(\mathbf{x}) \right) \in \mathbb{R}^n$$

the *gradient (vector) of f at $\mathbf{x}$* (we call ∇ “nabla.”). We denote the gradient of f at $\mathbf{x}$ by $\text{grad} f(\mathbf{x})$ or (simply) by $f'(\mathbf{x})$.

Partial Derivatives

Exercise 3.3 (Optional)/Example 11.2

Consider $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$\begin{aligned}\overbrace{f(x_1, x_2)}^{f(x)} &= \overbrace{(x_1 \ x_2)}^{x^T} \overbrace{\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}}^A \overbrace{\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}}^x \\ &= a_{11}x_1^2 + (a_{12} + a_{21})x_1x_2 + a_{22}x_2^2.\end{aligned}$$

Denote

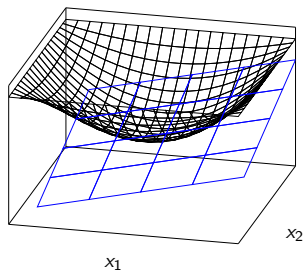
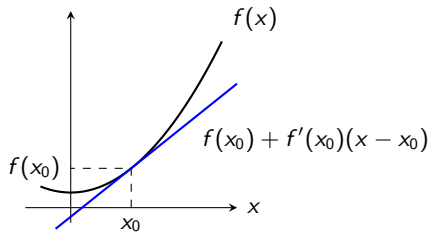
$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \text{ and } A^T = \begin{pmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{pmatrix}.$$

Show that

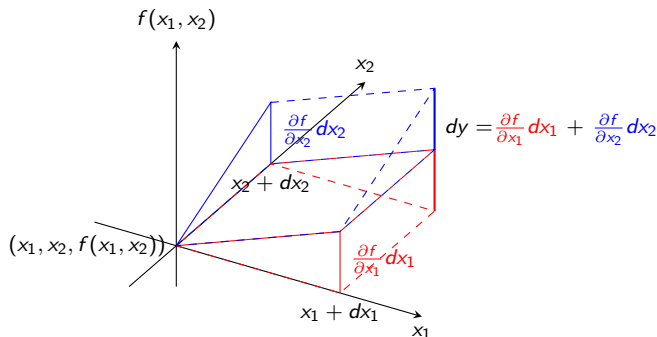
$$\nabla f(x_1, x_2) = \begin{pmatrix} 2a_{11} & a_{12} + a_{21} \\ a_{12} + a_{21} & 2a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = (A + A^T) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

Differentiation

The second basic idea is
to approximate a function $f(x_1, x_2, \dots, x_n)$ by a tangent plane.



Differentiation



$$f(x_1 + h_1, x_2 + h_2) - f(x_1, x_2) \approx \frac{\partial f(x_1, x_2)}{\partial x_1} h_1 + \frac{\partial f(x_1, x_2)}{\partial x_2} h_2$$

$$\Rightarrow dy = \frac{\partial f(x_1, x_2)}{\partial x_1} dx_1 + \frac{\partial f(x_1, x_2)}{\partial x_2} dx_2.$$

First-Order Condition

First-Order Condition

Let $f : A \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$.

Suppose that f is differentiable at an interior point x of A .

If x is a local extremal point (i.e., a local maximizer or a local minimizer), then

$$\underbrace{\nabla f(x)}_{\frac{\partial f(x)}{\partial x_i} = 0 \text{ for every } i} = \mathbf{0}.$$

Concave Maximization and Convex Minimization

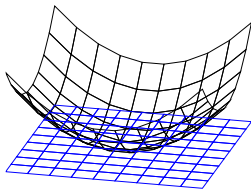
Let $C \subseteq \mathbb{R}^n$ be a convex set, and let $f : C \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ be a $\begin{cases} \text{concave} \\ \text{convex} \end{cases}$ function differentiable on $\text{int}(C)$ and continuous on C .

A point $x \in \text{int}(C)$ is a global $\begin{cases} \text{maximizer} \\ \text{minimizer} \end{cases}$ of f on C

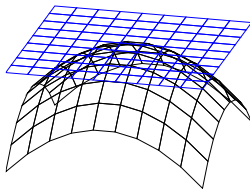
if and only if $\nabla f(x) = \mathbf{0}$.

First-Order Condition

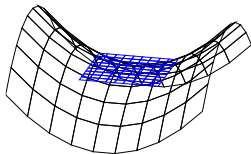
$$x^2 + y^2$$



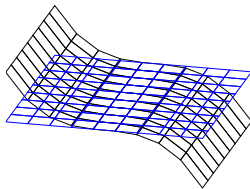
$$-x^2 - y^2$$



$$x^2 - y^2$$



$$-x^3$$



Representative Firm's Problem (Optional Exercise)

- Let $\Pi : \mathbb{R}_+^2 \rightarrow \mathbb{R}$ be defined by

$$\underbrace{\Pi(K, L)}_{\text{profit}} := AK^\alpha L^{1-\alpha} - (rK + wL).$$

- Consider:

$$\max_{K, L} \Pi(K, L).$$

- Denote by $Y := AK^\alpha L^{1-\alpha}$.
- The first-order conditions are:

$$\begin{aligned} \frac{\partial Y}{\partial K} - r &= 0, \text{ i.e., } r = \alpha A \left(\frac{K}{L} \right)^{\alpha-1} \text{ and} \\ \frac{\partial Y}{\partial L} - w &= 0, \text{ i.e., } w = (1 - \alpha) A \left(\frac{K}{L} \right)^\alpha. \end{aligned}$$

- Also, it is worth noting that the first-order condition with respect to K yields

$$\alpha = \frac{rK}{Y} \text{ and } 1 - \alpha = \frac{wL}{Y}.$$

- Consider the problem of finding the “best” straight line

$$y = \beta_1 + \beta_2 x,$$

or equivalently the best values $(\hat{\beta}_1, \hat{\beta}_2)$ of $(\beta_1, \beta_2) \in \mathbb{R}^2$, to fit a given set of data points $((x_i, y_i))_{i=1}^n$.

- In least squares regressions the best estimates $(\hat{\beta}_1, \hat{\beta}_2)$ are defined as values (β_1, β_2) solving

$$\min_{(\beta_1, \beta_2)} \underbrace{\sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i)^2}_{=: S(\beta_1, \beta_2)},$$

i.e., minimizing the **sum** of **squared deviations** of the observations y_i from the fitted values $\beta_1 + \beta_2 x_i$.

OLS: Calculus Approach I (Optional)

- Since this is a differentiable convex function, the first-order conditions with respect to β_1 and β_2 are, respectively,

$$\begin{aligned} n\hat{\beta}_1 + \left(\sum_{i=1}^n x_i\right) \hat{\beta}_2 &= \sum_{i=1}^n y_i, \text{ and} \\ \left(\sum_{i=1}^n x_i\right) \hat{\beta}_1 + \left(\sum_{i=1}^n x_i^2\right) \hat{\beta}_2 &= \sum_{i=1}^n x_i y_i. \end{aligned}$$

- After some manipulations, we can express the first-order conditions as

$$\begin{pmatrix} n & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{pmatrix} \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^n y_i \\ \sum_{i=1}^n x_i y_i \end{pmatrix}, \quad (1)$$

that is,

$$\begin{pmatrix} 1 & \frac{1}{n} \sum_{i=1}^n x_i \\ \frac{1}{n} \sum_{i=1}^n x_i & \frac{1}{n} \sum_{i=1}^n x_i^2 \end{pmatrix} \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \begin{pmatrix} \frac{1}{n} \sum_{i=1}^n y_i \\ \frac{1}{n} \sum_{i=1}^n x_i y_i \end{pmatrix}.$$

OLS: Calculus Approach I (Optional)

- Assuming the determinant of the leftmost matrix is not zero, that is, the (mathematical) variance $\frac{1}{n} \sum_{i=1}^n x_i^2 - \left(\frac{1}{n} \sum_{i=1}^n x_i\right)^2$ is not zero, the coefficients $(\hat{\beta}_1, \hat{\beta}_2)$ are

$$\begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \frac{1}{\frac{1}{n} \sum_{i=1}^n x_i^2 - \left(\frac{1}{n} \sum_{i=1}^n x_i\right)^2} \begin{pmatrix} \frac{1}{n} \sum_{i=1}^n x_i^2 & -\frac{1}{n} \sum_{i=1}^n x_i \\ -\frac{1}{n} \sum_{i=1}^n x_i & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{n} \sum_{i=1}^n y_i \\ \frac{1}{n} \sum_{i=1}^n x_i y_i \end{pmatrix} \\ = \frac{1}{\frac{1}{n} \sum_{i=1}^n x_i^2 - \left(\frac{1}{n} \sum_{i=1}^n x_i\right)^2} \left(\left(\frac{1}{n} \sum_{i=1}^n x_i^2\right) \left(\frac{1}{n} \sum_{i=1}^n y_i\right) - \left(\frac{1}{n} \sum_{i=1}^n x_i\right) \left(\frac{1}{n} \sum_{i=1}^n x_i y_i\right) \right).$$

- It can also be seen that

$$\hat{\beta}_2 = \frac{\sigma_{xy}}{\sigma_x^2} \text{ and } \hat{\beta}_1 = \mu_y - \hat{\beta}_2 \mu_x.$$

OLS: Calculus Approach II

Let

$$f(x) = \beta_1 + \beta_2 x = \begin{pmatrix} \beta_1 & \beta_2 \end{pmatrix} \begin{pmatrix} 1 \\ x \end{pmatrix}.$$

Then, the problem is:

$$\min_{(\beta_1, \beta_2)} \sum_{i=1}^n (y_i - f(x_i))^2.$$

The first-order conditions are

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^n (y_i - f(x_i)) \\ \sum_{i=1}^n (y_i - f(x_i)) x_i \end{pmatrix} = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \end{pmatrix} \begin{pmatrix} y_1 - f(x_1) \\ y_2 - f(x_2) \\ \vdots \\ y_n - f(x_n) \end{pmatrix}.$$

OLS: Calculus Approach II

Thus, we obtain

$$\begin{aligned} \overbrace{\begin{pmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \end{pmatrix}}^{X^T} \overbrace{\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}}^y &= \begin{pmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \end{pmatrix} \begin{pmatrix} f(x_1) \\ f(x_2) \\ \vdots \\ f(x_n) \end{pmatrix} \\ &= \underbrace{\begin{pmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \end{pmatrix}}_{X^T} \underbrace{\begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}}_X \underbrace{\begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix}}_{\hat{\beta}}. \end{aligned}$$

OLS: Calculus Approach II

Hence, we obtain

$$X^T \mathbf{y} = X^T X \hat{\boldsymbol{\beta}},$$

where

$$X = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}, \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \text{ and } \hat{\boldsymbol{\beta}} = \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix}.$$

Assuming that $X^T X$ is invertible, we obtain the formula

$$\boxed{\hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T \mathbf{y}.}$$

Aside: Projection Theorem

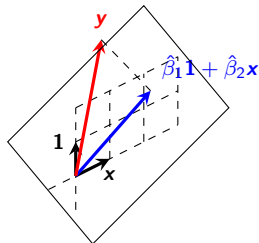
- Observe

$$S(\beta_1, \beta_2) = \|\mathbf{y} - \beta_1 \mathbf{1} - \beta_2 \mathbf{x}\|^2,$$

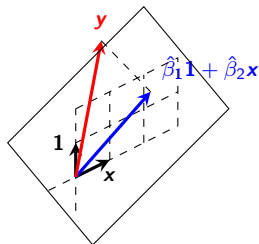
where $\mathbf{x} = (x_1, x_2, \dots, x_n)'$, $\mathbf{1} = (1, 1, \dots, 1)'$, and $\mathbf{y} = (y_1, y_2, \dots, y_n)'$.

- The coefficients $(\hat{\beta}_1, \hat{\beta}_2)$ are determined so that the vector $\mathbf{y} - \beta_1 \mathbf{1} - \beta_2 \mathbf{x}$ is perpendicular to the vector space spanned by $\mathbf{1}$ and $\mathbf{x}$:

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \end{pmatrix} \begin{pmatrix} y_1 - f(x_1) \\ y_2 - f(x_2) \\ \vdots \\ y_n - f(x_n) \end{pmatrix}.$$



Aside: Projection Theorem



- Recall:

$$\hat{\beta} = (X^T X)^{-1} X^T \mathbf{y}.$$

- Thus, the projection of $\mathbf{y}$ onto the space spanned by $\mathbf{1}$ and $\mathbf{x}$:

$$X \hat{\beta} = \underbrace{X(X^T X)^{-1} X^T}_{\text{projection matrix}} \mathbf{y}.$$

Example 8.1

- We simulate the set of observations $(x_i, y_i)_{i=1}^n$ and numerically find the coefficients $\hat{\beta} = (\hat{\beta}_1, \hat{\beta}_2)'$ to minimize

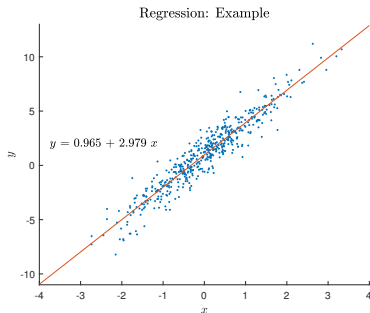
$$S(\beta_1, \beta_2) = \|\mathbf{y} - \beta_1 - \beta_2 \mathbf{x}\|^2.$$

- In the simulation, note that $\hat{\beta} = (X^T X)^{-1} X^T \mathbf{y}$ is also a random variable.
- Let $n = 500$. Let $\mathbf{x} = (x_i)_{i=1}^n$ be a vector of independent draws from the standard normal distribution. Let $\mathbf{y} = (y_i)_{i=1}^n$ be $y_i = 1 + 3x_i + \varepsilon_i$, where let $(\varepsilon_i)_{i=1}^n$ be a vector of independent draws from the standard normal distribution. Thus, we assume that the true parameter values are $(1, 3)$, although in reality we do not observe the true parameter values.

OLS

Example 8.1 (Cont'd)

- In the following simulation with MATLAB, we generate $\mathbf{x}$ and $\mathbf{y}$ (since we will fix the seed of the random number generation in the code, we will get the same result every time we run the code).
- Then, $\hat{\beta}$ is computed by the above projection formula (the coefficients are also compared with the MATLAB built-in function, regress).
- Then, we plot $(x_i, y_i)_{i=1}^n$ and $y = \hat{\beta}_1 + \hat{\beta}_2 x$.



Integration

- We use integration when, for instance, we compute the expectation or the variance of some random variable.
- With the aim of computing the expectation and variance of a normally distributed random variable, we review integration by parts and integration by substitution.

Question

- Suppose that X is normally distributed with mean μ and variance σ^2 . Its pdf is:

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

- We confirm that

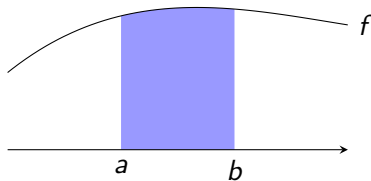
$$\mathbb{E}[X] = \mu \text{ and } \mathbb{V}[X] = \sigma^2.$$

Definite Integration

Consider a function $f : [a, b] \rightarrow \mathbb{R}$ and its (Riemann) integral:

$$\int_a^b f(x) dx.$$

When does it exist?



Proposition 3.11

- 1 If f is a continuous function, then it is (Riemann) integrable.
- 2 If f is a bounded function with a finite number of discontinuities, then it is (Riemann) integrable.
- 3 If f is a monotone function, then it is (Riemann) integrable.

Indefinite Integration

Definitions 3.7 and 3.9

Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function.

- ① The indefinite integral of f is a function $F : [a, b] \rightarrow \mathbb{R}$ defined by

$$F(x) := \int_a^x f(t)dt \text{ for all } x \in [a, b].$$

- ② An anti-derivative (a primitive function) of f is a function $F : [a, b] \rightarrow \mathbb{R}$ such that

$$F'(x) = f(x) \text{ for all } x \in (a, b).$$

- ③ We denote a generic anti-derivative of f by

$$\int f(x)dx = F(x) + c,$$

where (i) $F : [a, b] \rightarrow \mathbb{R}$ is an anti-derivative and (ii) c is a constant real number called the constant of integration.

Indefinite Integration

Fundamental Theorem of Calculus

- ① Let $f : [a, b] \rightarrow \mathbb{R}$ be a (Riemann) integrable function. The derivative of the indefinite integral is the original function f :

$$\frac{d}{dx} \left(\int f(x) dx \right) = f.$$

- ② Let $f : [a, b] \rightarrow \mathbb{R}$ be a differentiable function. The indefinite integral of the derivative is the original function plus the integration constant:

$$\int f'(x) dx = f + c.$$

$$f(x) = \frac{d}{dx} \left(\int_a^x f(t) dt \right) \qquad \int_a^x f(t) dt = F(x) - F(a)$$

Integration by Parts

- If f and g are differentiable, then so is its product and

$$(fg)' = f'g + fg'.$$

- So, we have

$$f'g = (fg)' - fg'.$$

- Then, we get

$$\begin{aligned}\int f'(x)g(x)dx &= \boxed{\int (fg)'(x)dx} - \int f(x)g'(x)dx \\ &= \boxed{f(x)g(x)} - \int f(x)g'(x)dx.\end{aligned}$$

Integration by Parts

Integration by Parts (Proposition 3.15)

Suppose that f and g are differentiable. Then:

$$\int f'(x)g(x)dx = f(x)g(x) - \int f(x)g'(x)dx.$$

Or:

$$\int_a^b f'(x)g(x)dx = [f(x)g(x)]_a^b - \int_a^b f(x)g'(x)dx.$$

Integration by Parts

Integration-by-Parts revisited: “Integration of $f \cdot g$ ”

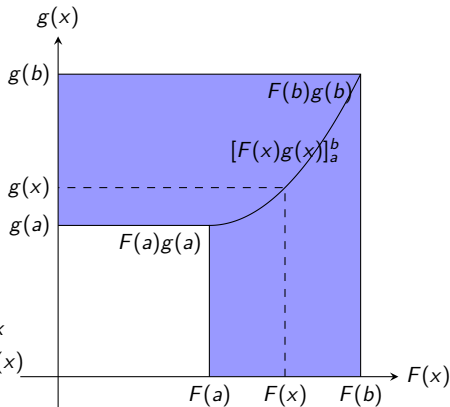
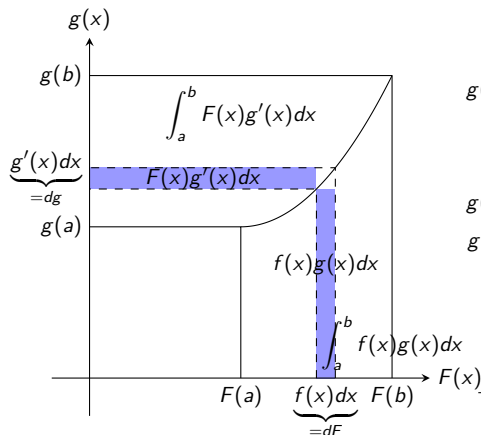
- Let F be an anti-derivative of f . If we take F instead of f , then

$$\int f(x)g(x)dx = F(x)g(x) - \int F(x)g'(x)dx.$$

- In this formula,
 - ① take an anti-derivative F of f and keep g ; then keep F and take the derivative g' .
 - ② Also, choose g so that Fg' becomes simple (needs some practice/experience).

Integration by Parts

Graphical Derivation (Optional)



$$\int_a^b f(x)g(x)dx = [F(x)g(x)]_a^b - \int_a^b F(x)g'(x)dx.$$

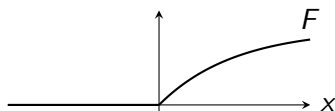
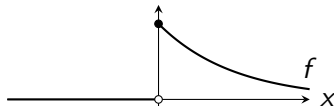
Integration by Parts

- The pdf of an exponential distribution is given by

$$f(x) = \begin{cases} 0 & \text{if } x < 0 \\ \lambda e^{-\lambda x} & \text{if } x \geq 0 \end{cases}.$$

- The cdf is given by

$$F(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ 1 - e^{-\lambda x} & \text{if } x > 0 \end{cases}.$$



Integration by Parts

Derivation of F (Optional)

- If $x \leq 0$, then

$$F(x) = \int_{-\infty}^x \underbrace{f(t)}_{=0} dt = 0.$$

- If $x > 0$, then

$$\begin{aligned} F(x) &= \int_{-\infty}^x f(t) dt \\ &= \int_0^x f(t) dt \\ &= \int_0^x \lambda e^{-\lambda t} dt \\ &= \left[-e^{-\lambda t} \right]_0^x \\ &= 1 - e^{-\lambda x}. \end{aligned}$$

Integration by Parts

Exponential Distribution

If $X \sim \text{Exp}(\lambda)$, then

$$\mathbb{E}[X] = \frac{1}{\lambda} \text{ and}$$
$$\mathbb{V}[X] = \frac{1}{\lambda^2}.$$

① We show:

$$\mathbb{E}[X] = \frac{1}{\lambda} \text{ and } \mathbb{E}[X^2] = \frac{2}{\lambda^2}.$$

② Then,

$$\begin{aligned}\mathbb{V}[X] &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\ &= \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2}.\end{aligned}$$

Integration by Parts

$$\begin{aligned}\mathbb{E}[X] &= \int_{-\infty}^{\infty} xf(x)dx \\&= \int_0^{\infty} xf(x)dx \\&= \int_0^{\infty} x(\lambda e^{-\lambda x})dx \\&= \int_0^{\infty} x(-e^{-\lambda x})'dx \\&= \left[x(-e^{-\lambda x}) \right]_0^{\infty} - \int_0^{\infty} (-e^{-\lambda x})dx \\&= \underbrace{\left[x(-e^{-\lambda x}) \right]_0^{\infty}}_{=0} + \int_0^{\infty} e^{-\lambda x}dx \\&= \int_0^{\infty} e^{-\lambda x}dx \\&= \left[-\frac{1}{\lambda} e^{-\lambda x} \right]_0^{\infty} \\&= \frac{1}{\lambda}.\end{aligned}$$

Integration by Parts

$$\begin{aligned}\mathbb{E}[X^2] &= \int_{-\infty}^{\infty} x^2 f(x) dx \\&= \int_0^{\infty} x^2 f(x) dx \\&= \int_0^{\infty} x^2 (\lambda e^{-\lambda x}) dx \\&= \int_0^{\infty} x^2 (-e^{-\lambda x})' dx \\&= \underbrace{\left[x^2 (-e^{-\lambda x}) \right]_0^{\infty}}_{=0} - \int_0^{\infty} 2x (-e^{-\lambda x}) dx \\&= 2 \int_0^{\infty} x e^{-\lambda x} dx \\&= \frac{2}{\lambda} \underbrace{\int_0^{\infty} x (\lambda e^{-\lambda x}) dx}_{=\mathbb{E}[X]=\frac{1}{\lambda}} = \frac{2}{\lambda^2}.\end{aligned}$$

Integration by Parts

Optional Exercise

- $\int \log x dx$

Since $(x \log x)' = \log x + 1$, we have

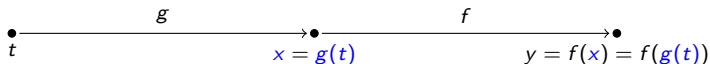
$$x \log x + c = \int \log x dx + x, \text{ i.e., } \int \log x dx = x(\log x - 1) + c.$$

By Integration by parts,

$$\begin{aligned} \int \log x dx &= \int \underbrace{1}_{\substack{f(x)=1 \\ F(x)=x}} \cdot \underbrace{\log x}_{\substack{g(x)=\log x \\ g'(x)=1/x}} dx \\ &= x \cdot \log x - \int x \frac{1}{x} dx = x \log x - x + c = x(\log x - 1) + c. \end{aligned}$$

Integration by Substitution

- Consider the composite of two functions:



- Specifically, consider $y = f(x) = x^4$ and $x = g(t) = 2t + 3$.
So, $f(g(t)) = (2t + 3)^4$.
- What is

$$\int (2t + 3)^4 dt?$$

- Can we utilize the fact that

$$\int x^4 dx = \frac{1}{5}x^5?$$

Integration by Substitution

- Let's check whether $\frac{1}{5}x^5 = \frac{1}{5}(2t+3)^5$ is an anti-derivative.
- It is not an anti-derivative because, by the chain rule,

$$\frac{d}{dt} \left(\frac{1}{5}(2t+3)^5 \right) = (2t+3)^4 \times \underbrace{2}_{=(2t+3)'}$$

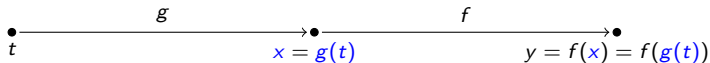
- It seems that the answer is

$$\int (2t+3)^4 dt = \frac{1}{2} \frac{1}{5} (2t+3)^5 + c = \frac{1}{10} (2t+3)^5 + c.$$

- How can we obtain the formula for the indefinite integral of a composite function by using the chain rule ?

Integration by Substitution

- Consider the two functions f and g :



- Suppose that g is differentiable while F is an anti-derivative of f .
- The indefinite integral of f satisfies

$$\begin{aligned}\int f(x) dx &= F(x) + c && (F: \text{anti-derivative}) \\ &= F(g(t)) + c && (x = g(t)) \\ &= \int (F(g(t)))' dt && \left(\int (F \circ g)'(t) dt = (F \circ g)(t) + c \right) \\ &= \int F'(g(t)) g'(t) dt && (\text{chain rule}) \\ &= \int f(g(t)) g'(t) dt. && (f = F')\end{aligned}$$

- Heuristically, since $x = g(t)$, we have $\frac{dx}{dt} = g'(t)$, so that $dx = g'(t) dt$.

Integration by Substitution

Proposition 3.16

Let $I \subseteq \mathbb{R}$ be an interval. Let $f : I \rightarrow \mathbb{R}$ and $g : [a, b] \rightarrow I$ be functions. Suppose further that f is (Riemann) integrable and g is differentiable on (a, b) and continuous on $[a, b]$.

① For the indefinite integral,

$$\int f(\textcolor{red}{x}) \textcolor{blue}{dx} = \int f(\textcolor{red}{g}(t)) \textcolor{blue}{g}'(t) dt.$$

② For the definite integral

$$\int_{g(a)}^{g(b)} f(\textcolor{red}{x}) \textcolor{blue}{dx} = \int_a^b f(\textcolor{red}{g}(t)) \textcolor{blue}{g}'(t) dt.$$

t	$a \rightarrow b$
$x = g(t)$	$g(a) \rightarrow g(b)$

Integration by Substitution

- $\int_1^2 (2t + 3)^4 dt.$
- Let $x = 2t + 3$ so that $\frac{dx}{dt} = 2$. Thus, $dx = 2dt$ or $dt = \frac{1}{2}dx$.
- Also, when t ranges from 1 to 2, x ranges from 5 to 7.

$$\begin{array}{c|c} t & 1 \rightarrow 2 \\ \hline x = 2t + 3 & 5 \rightarrow 7 \end{array}$$

- Then,

$$\begin{aligned} \int_1^2 (2t + 3)^4 dt &= \int_5^7 x^4 \frac{1}{2} dx \\ &= \frac{1}{2} \int_5^7 x^4 dx \\ &= \frac{1}{2} \left[\frac{1}{5} x^5 \right]_5^7 = \frac{1}{10} (7^5 - 5^5) = \frac{6841}{10}. \end{aligned}$$

Integration by Substitution

Optional Exercise (close to what we will do)

- $\int x e^{-x^2} dx$

Let $y = -x^2$ so that $\frac{dy}{dx} = -2x$, i.e., $dy = -2dx$. Then,

$$\begin{aligned}\int x e^{-x^2} dx &= \int e^y \left(-\frac{1}{2} \right) dy \\ &= -\frac{1}{2} \int e^y dy \\ &= -\frac{1}{2} e^y + c = -\frac{1}{2} e^{-x^2} + c.\end{aligned}$$

Normal Distribution

Expectation

$$\begin{aligned}\mathbb{E}[X] &= \int_{-\infty}^{\infty} xf(x)dx \\&= \int_{-\infty}^{\infty} (x - \mu)f(x)dx + \mu \\&= \int_{-\infty}^{\infty} (x - \mu) \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) dx + \mu \\&= \int_{-\infty}^{\infty} \frac{x - \mu}{\sigma} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x - \mu}{\sigma}\right)^2\right) dx + \mu.\end{aligned}$$

Normal Distribution

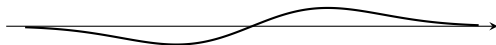
Expectation

Letting $t = \frac{x-\mu}{\sigma}$, we have $dt = \frac{dx}{\sigma}$, i.e., $dx = \sigma dt$.

Also, t ranges from $-\infty$ to ∞ . Then:

$$\begin{aligned}\mathbb{E}[X] &= \int_{-\infty}^{\infty} \frac{x-\mu}{\sigma} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{x-\mu}{\sigma}\right)^2\right) dx + \mu \\&= \mu + \int_{-\infty}^{\infty} t \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) \sigma dt \\&= \mu + 0 = \mu,\end{aligned}$$

as the function $t \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right)$ is an odd function.



Normal Distribution

Expectation

We can also confirm

$$\begin{aligned}& \int_{-\infty}^{\infty} (x - \mu) \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) dx \\&= \int_{-\infty}^{\infty} (-\sigma^2) \left[\frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) \right]' dx \\&= (-\sigma^2) \left[\frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) \right]_{-\infty}^{\infty} \\&= 0 - 0 = 0.\end{aligned}$$

Normal Distribution

Variance

$$\begin{aligned}\mathbb{V}[X] &= \int_{-\infty}^{\infty} (x - \mu)^2 \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) dx \\&= \int_{-\infty}^{\infty} (x - \mu) \left(\frac{(-\sigma^2)}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) \right)' dx \\&= \left[(x - \mu) \frac{(-\sigma^2)}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) \right]_{-\infty}^{\infty} + \underbrace{\sigma^2 \int_{-\infty}^{\infty} \left(\frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) \right) dx}_{=1} \\&= \sigma^2.\end{aligned}$$

Normal Distribution

Gaussian Integral (Optional)

- Here, we prove

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}.$$

- More generally, we have

$$\int_{-\infty}^{\infty} e^{-a(x+b)^2} dx = \sqrt{\frac{\pi}{a}}.$$

- This is because, by letting $x = \frac{y}{\sqrt{a}} - b$, we have

$$\int_{-\infty}^{\infty} e^{-a(x+b)^2} dx = \int_{-\infty}^{\infty} e^{y^2} \frac{dy}{\sqrt{a}} = \sqrt{\frac{\pi}{a}}.$$

- Hence, (letting $a = \frac{1}{2\sigma^2}$ and $b = -\mu$) we obtain

$$\frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = 1.$$

Normal Distribution

Gaussian Integral (Optional)

- Define

$$I := \int_{-\infty}^{\infty} e^{-x^2} dx.$$

- We consider:

$$\begin{aligned} I^2 &= \left(\int_{-\infty}^{\infty} e^{-x^2} dx \right) \left(\int_{-\infty}^{\infty} e^{-y^2} dy \right) \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dy dx. \end{aligned}$$

- Thus, we show that $I^2 = \pi$.

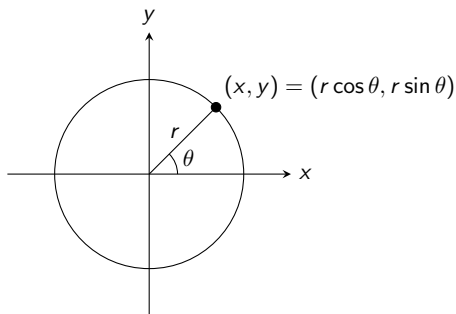
Normal Distribution

Gaussian Integral (Optional)

- Define

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} r \cos \theta \\ r \sin \theta \end{pmatrix},$$

where $r > 0$ and $\theta \in [0, 2\pi)$.



Normal Distribution

Gaussian Integral (Optional)

- Since the determinant of the Jacobi matrix is

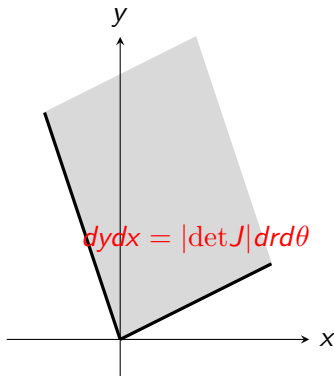
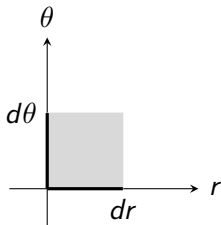
$$\begin{aligned}\det J &= \det \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix} \\ &= \det \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix} \\ &= r(\cos^2 \theta + \sin^2 \theta) = r(> 0)\end{aligned}$$

and since $x^2 + y^2 = r^2$, it follows that

$$\begin{aligned}I^2 &= \int_0^{2\pi} \int_0^\infty \underbrace{e^{-r^2}}_{=e^{-(x^2+y^2)}=dydx} \underbrace{rdrd\theta}_{=dydx} \\ &= 2\pi \left[-\frac{1}{2}e^{-r^2} \right]_0^\infty = \pi.\end{aligned}$$

Normal Distribution

Gaussian Integral (Optional)



- When $x = g(t)$, $dx = g'(t)dt$: $g'(t)$ is the rate at which a change in t (i.e., dt) is enlarged to the change in x (i.e., dx).
- When $x = x(r, \theta)$ and $y = y(r, \theta)$, $|\det J|$ is the rate at which the $drd\theta$ is enlarged to $dx dy$.

Leibniz Formula (Optional)

Problem (Example 3.22)

Define the continuation value at time t as:

$$V(t) := \int_t^{\infty} e^{-\rho(s-t)} f(s) ds,$$

where $f : \mathbb{R} \rightarrow \mathbb{R}$ is a bounded continuous function. Show that

$$V'(t) = \underbrace{\rho V(t)}_{\text{Discounted today's value}} - \underbrace{f(t)}_{\text{value}}.$$

- 1 One way is to compute the limit of $\frac{V(t+\Delta t) - V(t)}{\Delta t}$.
- 2 How can we take the derivative of an integral with varying endpoints?

Leibniz Formula

Theorem 3.3 (Leibniz Formula)

Let $f : [a, b] \times [\underline{r}, \bar{r}] \rightarrow \mathbb{R}$ be a function with the following properties: (i) $f(\cdot, r)$ is continuous in $[a, b]$ for every $r \in [\underline{r}, \bar{r}]$; and (ii) $\frac{\partial}{\partial r} f(x, r)$ exists and is continuous for every $(x, r) \in [a, b] \times [\underline{r}, \bar{r}]$. Let $A, B : [\underline{r}, \bar{r}] \rightarrow [a, b]$ be continuously differentiable functions. Define

$$V(r) := \int_{A(r)}^{B(r)} f(x, r) dx.$$

Then,

$$V'(r) = f(B(r), r)B'(r) - f(A(r), r)A'(r) + \int_{A(r)}^{B(r)} \frac{\partial}{\partial r} f(x, r) dx. \quad (2)$$

Leibniz Formula

Exercise 3.6

This exercise guide you to (informally) prove Theorem 3.3.

- 1 Let $g : [a, b] \times [\underline{r}, \bar{r}] \rightarrow \mathbb{R}$ be such that, for each $r \in [\underline{r}, \bar{r}]$, $g(\cdot, r)$ is a primitive function of $f(\cdot, r)$. That is,

$$\frac{\partial}{\partial x} g(x, r) = f(x, r).$$

Now, express V using g .

- 2 Derive Expression (2).
(Note: this part calls for the chain rule for a multi-variate function.)

- Instead, we will derive the Leibniz formula more intuitively.

Leibniz Formula

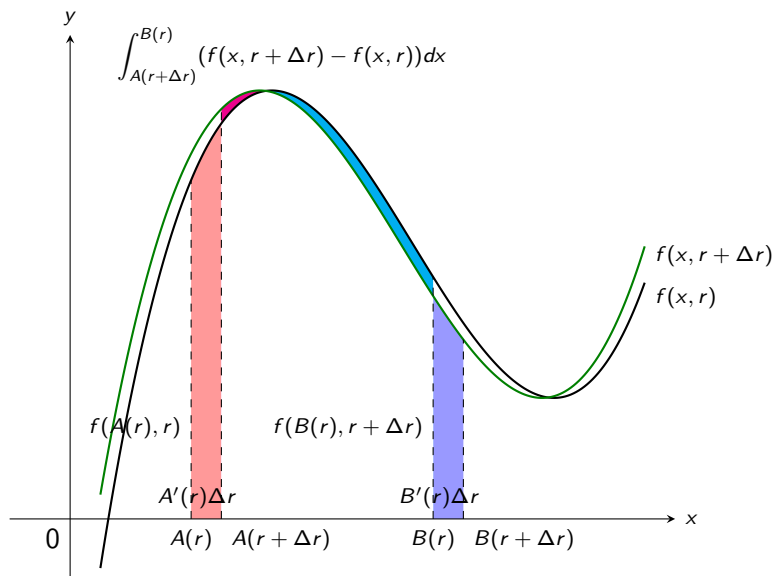


Figure: Illustration of Expression (2)

Leibniz Formula

Intuitive Derivation

For two points r and $r + \Delta r$ where Δr is small, the figure illustrates that the difference between

$$V(r + \Delta r) = \int_{A(r+\Delta r)}^{B(r+\Delta r)} f(x, r + \Delta r) dx \text{ and } V(r) = \int_{A(r)}^{B(r)} f(x, r) dx$$

is approximately

$$B'(r)\Delta r f(B(r), r + \Delta r) - A'(r)\Delta r f(x, r) + \int_{A(r+\Delta r)}^{B(r)} (f(x, r + \Delta r) - f(x, r)) dx.$$

Thus, the difference quotient $\frac{V(r+\Delta r) - V(r)}{\Delta r}$ is approximately

$$B'(r)f(B(r), r + \Delta r) - A'(r)f(x, r) + \int_{A(r+\Delta r)}^{B(r)} \frac{f(x, r + \Delta r) - f(x, r)}{\Delta r} dx.$$

Letting $\Delta r \rightarrow 0$ yields Expression (2):

$$B'(r)f(B(r), r) - A'(r)f(x, r) + \int_{A(r)}^{B(r)} \frac{\partial f(x, r)}{\partial r} dx.$$